

Functions, Injectivity, Surjectivity, Bijections

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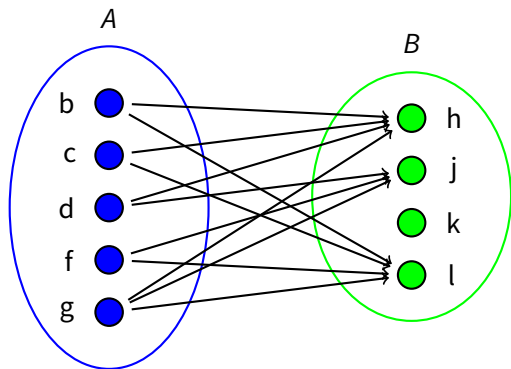
Overview

1 Relation Diagrams (4.4.1)

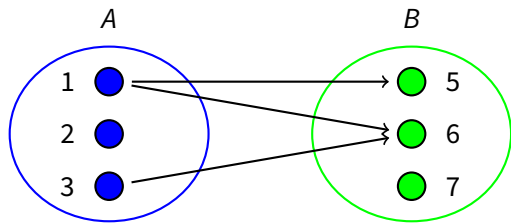
2 Relational Images (4.4.2)

Binary relations

Definition. A *binary relation*, R , consists of a set, A , called the domain of R , a set, B , called the codomain of R , and a subset of $A \times B$ called the graph of R .



Properties of relations

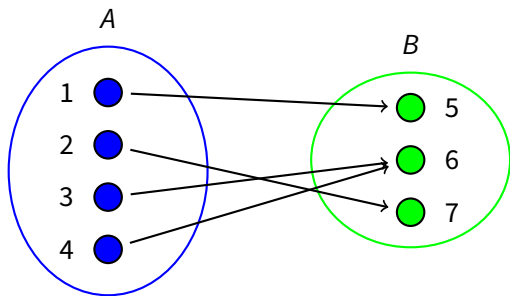


A binary relation:

- is a *partial function* when it has the $[\leq 1 \text{ arrow out}]$ property.
Book: “function”. Us: “function” is $[= 1 \text{ arrow out}]$ property.
- is *surjective* when it has the $[\geq 1 \text{ arrows in}]$ property.
- is *total* when it has the $[\geq 1 \text{ arrows out}]$ property.
- is *injective* when it has the $[\leq 1 \text{ arrow in}]$ property.
- is *bijective* when it has both the $[= 1 \text{ arrow out}]$ and the $[= 1 \text{ arrow in}]$ properties.

Example relation #1

partial function: $[\leq 1 \text{ out}]$. surjective: $[\geq 1 \text{ in}]$. total: $[\geq 1 \text{ out}]$. injective: $[\leq 1 \text{ in}]$.
 bijective: $[= 1 \text{ out}]$ and $[= 1 \text{ in}]$.

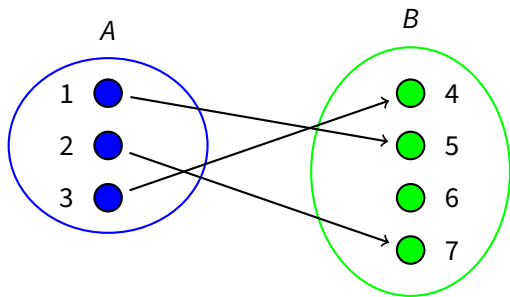


Partial function; surjective; total. Not injective, not bijective.

Summary: a surjective function. (Implies partial function and total.)

Example relation #2

partial function: $[\leq 1 \text{ out}]$. surjective: $[\geq 1 \text{ in}]$. total: $[\geq 1 \text{ out}]$. injective: $[\leq 1 \text{ in}]$.
 bijective: $[= 1 \text{ out}]$ and $[= 1 \text{ in}]$.



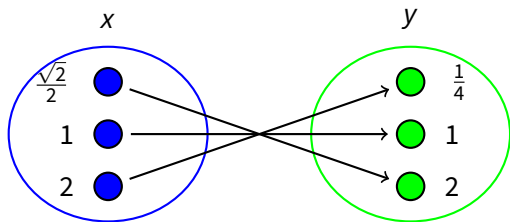
Partial function; total; injective. Not surjective, not bijective.

Summary: an injective function. (Implies partial function and total.)

Example relation #3

partial function: $[\leq 1 \text{ out}]$. surjective: $[\geq 1 \text{ in}]$. total: $[\geq 1 \text{ out}]$. injective: $[\leq 1 \text{ in}]$.
 bijective: $[= 1 \text{ out}]$ and $[= 1 \text{ in}]$.

Equation $y = 1/x^2$ on $\mathbb{R}^+$. x is an element in the domain, y is an element in the co-domain.



Partial function; surjective; total; injective; bijective.

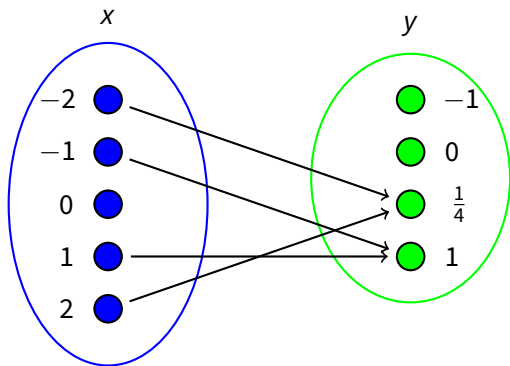
Summary: a bijective (partial) function. (Implies everything else.)

Example relation #4

partial function: $[\leq 1 \text{ out}]$. surjective: $[\geq 1 \text{ in}]$. total: $[\geq 1 \text{ out}]$. injective: $[\leq 1 \text{ in}]$.

bijective: $[= 1 \text{ out}]$ and $[= 1 \text{ in}]$.

Equation $y = 1/x^2$ on $\mathbb{R}$.



Partial function. Not anything else.

Image definition

Definition. The *image* of a set $Y \subseteq A$ under a relation $R : A \rightarrow B$, written $R(Y)$, is the subset of elements of the codomain B of R that are related to some element in Y .

In terms of the relation diagram, $R(Y)$ is the set of points with an arrow coming in that starts from some point in Y .

$$R(Y) = \{x \in B \mid \exists y \in Y, y R x\}.$$

Inverse definition

Definition: The *inverse* R^{-1} of a relation $R : A \rightarrow B$ is the relation from B to A defined by the rule

$$b R^{-1} a \leftrightarrow a R b.$$

Definition: The image of a set under the relation R^{-1} is called the *inverse image* of the set. That is, the inverse image of a set X under the relation R is defined to be $R^{-1}(X)$.

Example: $x R y$ iff there's a dictionary word with first letter x and second letter y . The image $R(\{c, k\})$ is the letters that can appear after 'c' or 'k' at the beginning of a word. It's the set $\{a, e, h, i, l, n, o, r, u, v, w, y, z\}$.

The inverse image $R^{-1}(\{c, k\})$ is the letters that can appear before 'c' or 'k' at the beginning of a word. It's the set $\{a, e, i, o, s, t, u\}$.

Inverses of relations

What can we infer about R^{-1} if R is:

- partial function? injective
- surjective? total
- total? surjective
- injective? partial function
- bijective? bijective
- function? injective and surjective

More examples to consider

Make natural examples for each combination of properties.

- $y = \sqrt{x}$ on $\mathbb{R}$
- $y = \sqrt{16 - \sqrt{x}}$ on $\mathbb{R}$
- $y = |x + 10|$ on $\mathbb{Z}$
- $y = |x \bmod 2|$ on $\mathbb{Z}$
- $y = \sin(x)$ on $\mathbb{R}$
- $x^2 + y^2 = 10$ on $\mathbb{R}$